# Sai Chandra Mouli Kaki

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## Summary

B.Tech graduate from IIIT Nagpur with hands-on experience building **AI, computer vision, and edge computing systems**. Experienced in **real-time video analytics, deep learning model training, dataset annotation**, and deploying AI workloads on **NVIDIA Jetson** platforms. Skilled in Python, PyTorch, TensorFlow, OpenCV, DeepStream, GStreamer, CUDA, and Docker, with a focus on practical real-time AI applications.

## Education

Indian Institute of Information Technology, Nagpur (2026) — CGPA: 6.46
Bachelor of Technology in Electronics and Communication (IoT)

Tirumala Junior College (2021) — 96.8%

Tirumala High School (2019) — 10 CGPA

## Experience

**Streamux** | *AI Engineer Intern* — Nagpur, Maharashtra | March 2026 – Present

- Developed **real-time CCTV video analytics pipelines** using NVIDIA DeepStream for **object detection, tracking, counting**, and perimeter-based event detection on NVIDIA Jetson edge devices.
- Designed and optimized **GStreamer pipelines** for multi-stream video ingestion, preprocessing, inference, and post-processing, leveraging **CUDA-accelerated components** for real-time performance.
- Trained **custom multi-class classification models** by preparing and annotating datasets, organizing classes, configuring training pipelines, evaluating model performance, and iterating based on inference results.
- Integrated **computer vision inference workflows** into a Smart Surveillance VMS, processing live camera streams and generating event-driven AI outputs for automated surveillance use cases.
- Configured and deployed **NVIDIA Jetson Orin Nano** systems with Aetina carrier boards, setting up the software stack and AI inference environment for edge deployments.

## Projects

**Smart Traffic Monitoring System** | *YOLO, PyTorch, Computer Vision* — GitHub

- Engineered a **real-time traffic monitoring system** using **YOLO and PyTorch** for vehicle detection, tracking, traffic data extraction, and automated alert generation.
- Designed a complete **traffic scene analysis workflow** to reliably capture vehicle data under varying traffic densities and dynamic road conditions.

**Traffic Signal Optimization** | *SUMO, Reinforcement Learning, Python* — GitHub

- Built an **adaptive traffic signal optimization system** using **SUMO and Reinforcement Learning** to optimize signal decisions based on simulated traffic conditions.
- Evaluated different **signal-control strategies** using traffic flow and simulation-based performance metrics to compare adaptive decision-making approaches.

**Brain Tumor Segmentation** | *TensorFlow, OpenCV, Image Processing* — Kaggle

- Developed a **deep learning image segmentation pipeline** using **TensorFlow and OpenCV** for automated brain tumor segmentation.
- Evaluated segmentation quality using **Intersection over Union (IoU)** and iterated on the model and image-processing workflow to improve segmentation performance.

**Potato Disease Classification** | *Python, Deep Learning* — GitHub

- Developed an **end-to-end deep learning classification system** for identifying potato diseases from leaf images, covering model training, inference, and backend processing.
- Implemented a **low-latency demonstration workflow** for testing model predictions and validating the complete application pipeline.

## Skills

- **Programming:** Python, C/C++, SQL
- **AI/ML:** Deep Learning, Computer Vision, Object Detection, Object Tracking, Classification, Segmentation, Model Training, Dataset Annotation, Model Evaluation
- **Edge AI:** NVIDIA Jetson Orin Nano, DeepStream, GStreamer, CUDA, Docker, Aetina Carrier Boards, Real-time Systems
- **Frameworks/Tools:** PyTorch, TensorFlow, OpenCV, Git/GitHub, VS Code